

### Theoretical Questions:

#### Question 1: What is a Calculated Field and why is it important in business dashboards?

- **Definition:** A Calculated Field is a new data column created within Tableau using existing data from the source. It allows to perform mathematical operations, logical comparisons, or string manipulations without altering the original dataset.
- **Importance: Custom Metrics:** It enables the creation of business-specific KPIs (like "Profit Margin" or "Delivery Status") that may not exist in the raw data.
  - **Data Cleaning:** It can be used to group data or fix inconsistencies (e.g., renaming "N.Y." and "New York" to a single value).
  - **Dynamic Analysis:** It allows for logical segments (e.g., "High Value" vs. "Low Value" customers) which are essential for targeted business decisions.

#### Question 2 : Why can't parameters change visuals directly in Tableau?

- **The Nature of Parameters:** Parameters are static, "global" values (like a constant number or a string). They act as a container for a value provided by the user.
- **The Dependency:** A parameter does nothing by itself because it is not connected to the data source. To make a parameter "work," it must be referenced inside a **Calculated Field** or a **Filter**. The visual only changes when that calculation or filter updates based on the value currently stored in the parameter.

#### Question 3 : When should EXCLUDE LOD be used?

- **Usage Scenario:** EXCLUDE LOD expressions should be used when we want to calculate a value at a *higher* (less granular) level than what is currently shown in view.
- **Example:** If your view shows Sales by "Region" and "City," but you want to compare each city's sales against the total "Region" sales, you would use EXCLUDE [City] to ignore the city-level detail for that specific calculation.

#### Question 4 : Difference between FIXED and INCLUDE LOD expressions?

- **FIXED LOD:** Computes a value using only the specified dimensions, completely independent of whatever dimensions are in the view. It also ignores most filters unless they are "Added to Context."
- **INCLUDE LOD:** Computes a value using the specified dimensions *in addition* to whatever dimensions are already in the view. It is sensitive to filters and the level of detail currently on the rows/columns shelf.